

Learning goals:

- Understand and apply common steps along an ERP decoding pipeline;
- Understand ERPs and their spatial and temporal components;
- Understand temporal, spatial, and spatio-temporal decoding;
- Understand significance testing using the permutation test;
- Get familiar with the MNE toolbox for EEG analysis;
- Get familiar with sklearn for EEG decoding.

Question 1: Refresh your understanding

Refresh your understanding in the following concepts:

- EEG preprocessing: EEGLab series
- AUC metric: sklearn's tutorial, Wikipedia

Question 2: EEG decoding pipeline

- (a) Write down each of the steps necessary to go from a raw EEG recording that produces a long continuous EEG file, all the way to predicted classes as output by a classification model.
- (b) At each step of this pipeline, note down the dimensionality of the data.

Question 3: Evaluation metrics

- (a) Why is the classification accuracy not an appropriate metric for a dataset with unbalanced labels?
- (b) Name some metrics that can evaluate a decoding pipeline.
- (c) Name some metrics that can evaluate an application (not the decoding pipeline).

Question 4: Evaluation protocols

- (a) What is cross-validation used for?
- (b) Does cross-validation *prevent* overfitting?
- (c) What is *nested* cross-validation used for?

Question 5: Jupyter notebook ERP analysis

Now put the theory into practice by going through the ERP analysis Jupyter notebook.

Question 6: Submitting to Brightspace

For this assignment, please submit the following:

- The Jupyter notebook with your solutions
- A report (pdf extension) with the answers to the theoretical questions. In this report, add a line stating who contributed to what (please, adhere to the code of conduct). Additionally, please add a line on whether, where, and how you used GenAI for this assignment.

Question 7: (Optional) Test your understanding with real exam questions

- (a) ERP analysis
 - 2020-2021 resit Part 4
 - 2021-2022 resit Part 5
 - 2022-2023 Part 3
 - 2023-2024 Part 5
 - 2024-2025 Part 5